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FYP Project Storyline – Expected Final Result

Project Overview

The aim of my Final Year Project is to develop a procedural building generation tool inside Blender that helps 3D artists create buildings quickly and efficiently, without having to model every part manually.

In traditional 3D workflows, creating a building involves modelling walls, floors, windows, doors, and roofs individually. This process can take a significant amount of time, especially when an artist needs to create an entire city or environment. Making changes to existing buildings can also require a lot of additional work.

My project aims to solve this problem by creating a system that automatically generates buildings using predefined rules and modular 3D assets, but also to be able to get enough manual control. The user will be able to generate a building and control its appearance through adjustable parameters.

How the Project Will Work

The tool will be developed using Blender Geometry Nodes and Python scripting. The user will be able to adjust properties such as building width, height, number of floors, and other structural details through a simple interface.

Whenever a parameter changes, the building will update automatically, allowing the user to experiment with different designs without starting from scratch. A seed-based randomisation system will also allow users to generate different building variations from the same setup.

The main focus will be on making the tool practical, easy to use, and efficient within Blender. Additional features, such as viewport optimisation and support for custom modular assets will be implemented.

Expected Final Result

By the end of the project, I expect to deliver a working procedural building generator that can create complete customisable or can aid in creating 3D buildings fast within Blender. The final demonstration will show how a single generator can produce buildings with different dimensions and appearances, update them through parameter changes, and create multiple variations for an environment.

The project will be evaluated through functional self-testing, visual results, and a comparison of the time and effort required against traditional manual modelling.

Ultimately, the goal is to demonstrate how procedural modelling can reduce repetitive work, speed up environment creation, and give artists more time to focus on creativity and design.